IT602 FINALS NOTES

1. Network Virtualization

- Network Virtualization (NV) refers to abstracting network resources that were traditionally delivered in hardware to software. NV can combine multiple physical networks to one virtual, software-based network, or it can divide one physical network into separate, independent virtual networks.

Key Features:

- **Decouples network services from hardware:** Network functionalities like routing, switching, and firewalling are delivered in software.
- **Allows programmatic network management:** Administrators can create, configure, and manage virtual networks through software.
- **Enables flexibility and scalability:** Virtual networks can be adjusted easily without modifying the physical infrastructure.
- **Enhances security and efficiency:** Virtualized networks allow better control over data traffic and resource utilization.

□ **Example:** Cloud environments where multiple tenants share the same physical network but have separate, secure virtual networks through VLANs or VXLANs.

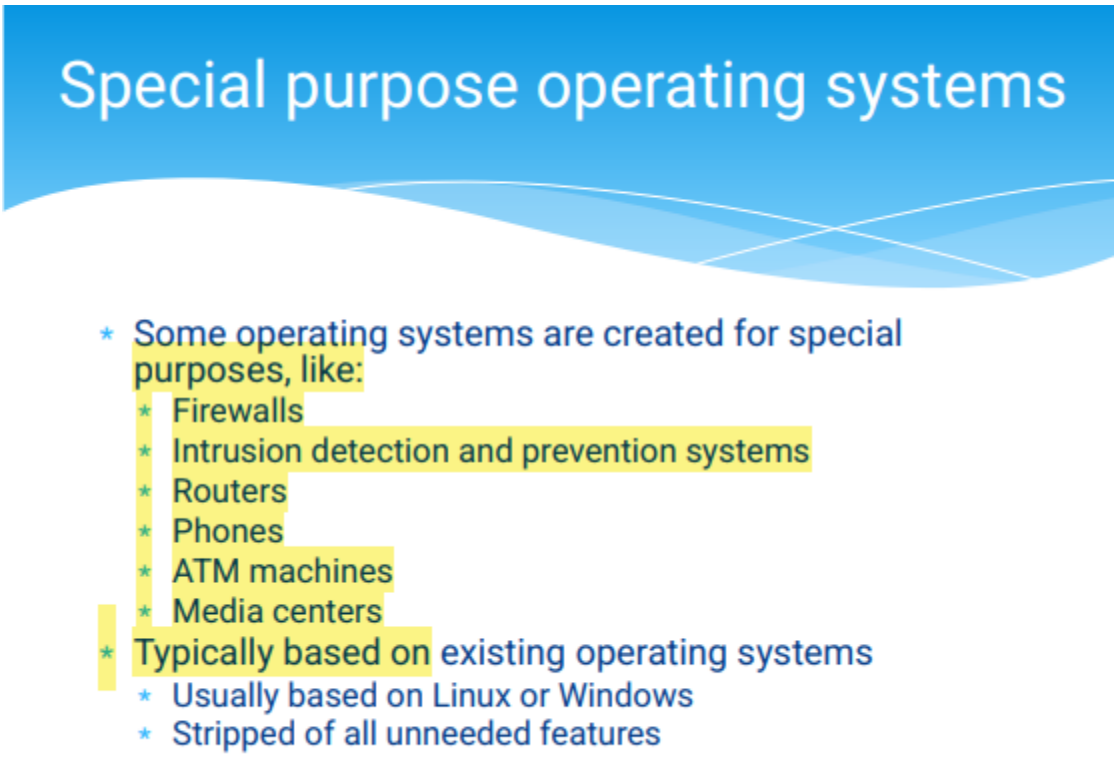
2. Container Technology

Container Technology is a type of server virtualization that allows multiple isolated environments (containers) to run on the same operating system kernel. Unlike traditional virtual machines, containers do not require a separate OS installation for each instance, making them lightweight and efficient.

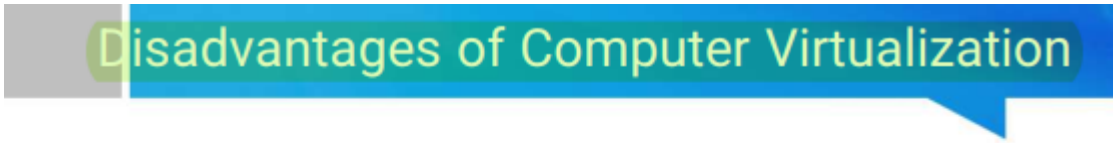
Key Features:

- **Isolation:** Each container runs independently with its own libraries and dependencies.
- **Portability:** Containers can run consistently across different environments, such as on-premises servers and cloud platforms.
- **Fast Deployment:** Applications in containers can be deployed quickly, making them ideal for DevOps and microservices architectures.

3. Special Purpose Operating System Names



DISADVANTAGES OF VIRTUALIZATION



- Because creating a new virtual machine is so easy, virtual machines tend to get created for all kinds of reasons
  - This effect is known as "virtual machine sprawl"
  - All VMs:
    - Must be managed
    - Use resources of the physical machine
    - Use power and cooling
    - Must be back-upped
    - Must be kept up to date by installing patches

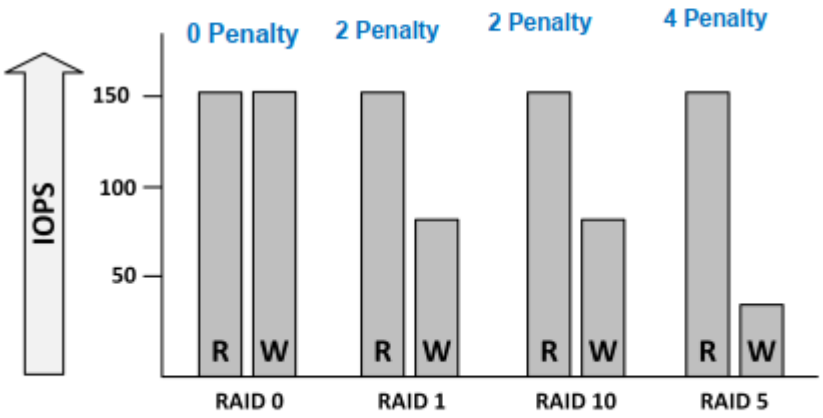
Disadvantage of virtualization

5. What is Vendor Lock-in?

Vendor lock-in occurs when a company becomes dependent on a particular vendor's technology, making it difficult to switch to another provider due to compatibility issues, high switching costs, or proprietary technologies.

RAID Penalty

- In RAID sets multiple disks are used to form one virtual disk (LUN)
- Writing data on multiple disks introduces some delay, known as the RAID penalty



Transparency and Accountability

Agar ek app user ka location data le rahi hai, to usay user ko clearly batana chahiye ke ye data kahan use hoga, aur agar kisi hacker ne data chura liya to app wale zimmedar honge usko recover karne ke liye.

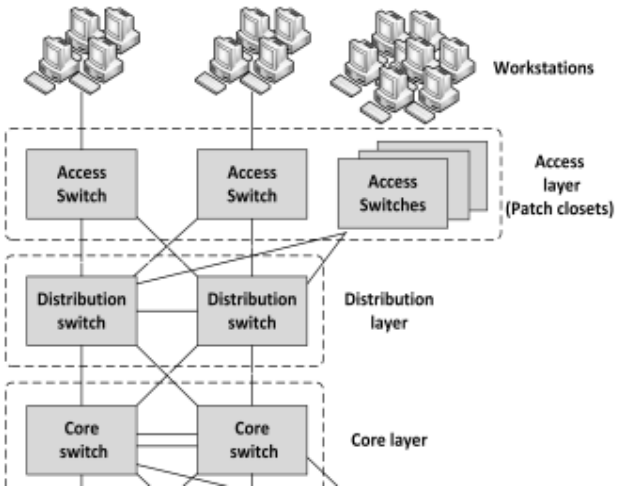
Short question

- Transparency in IT infrastructure practices involves being open about policies, procedures, and data usage. It also includes taking responsibility for any errors or security breaches that occur and working to rectify them promptly.

- Key points to emphasize:
  - Clear Data Policies
  - Data Usage Notifications
  - Data Sharing and Third-Party Disclosures
  - Regular Transparency Reports
  - Accountability Mechanisms
  - User Access to Data
  - Incident Reporting
  - User Education
  - Ethical Data Use for Research

Layered Network Topology

- A network infrastructure should be built up in layers
  - Improve availability and performance
  - Provides scalability
  - Provides deterministic routing
  - Avoids unmanaged ad-hoc data streams



Layered Network Topology

Long question

- Core layer
  - This is the center of the network
- Distribution layer
  - An intermediate layer between the core layer in the datacenter and the access switches in the patch closets
  - Combines the access layer data and sends its combined data to one or two ports on the core switches
- Access layer
  - Connect workstations and servers to the distribution layer
  - For servers, located at the top of the individual server racks or in blade enclosures
  - For workstations, placed in patch closets in various parts of the building

9. **READ Command in OS**

The **READ** command in an operating system is used to retrieve data from storage devices such as hard drives or memory.

Functions of the READ Command:

- Allows programs to access and retrieve data from files.
- Works with system calls like `read()` in UNIX/Linux.
- Ensures efficient data transfer between memory and storage devices.
- Can be synchronous (blocking) or asynchronous (non-blocking).

- Example:

```
int read(int handle, void *buffer, int
nbyte);
```

translates into the system call

```
READ(FILEHANDLE, DESTINATION DATA POINTER,
NUMBER_OF_BYTES)
```

Ethics for IT infrastructure

Ethics for IT infrastructure involves considering the moral and societal implications of the design, implementation, and use of technology that underpins an organization's information technology systems. Here are some key ethical considerations for IT infrastructure:

- Privacy and Data Protection
- Transparency and Accountability
- Accessibility
- Security and Cybersecurity
- Sustainability and Environmental Impact
- Fairness and Bias Mitigation
- Intellectual Property
- Responsibility in AI and Automation
- Long-Term Planning and Scalability
- Human-Centered Design
- Ethical Decision-Making
- Compliance and Legal Adherence

Long question

be accessed by (users on) other systems

– File sharing protocols:
  - NFS: originates from UNIX
  - SMB/CIFS: originates from Windows

Short question

Short question

- Systems management console
  - The systems management console connects to all hypervisors and virtual machines
  - When the security of the systems management console is breached, security is breached on all virtual machines
  - Not all systems managers should have access to all virtual machines
  - Special user accounts and passwords should be configured for high risk operations like shutting down physical machines or virtualized clusters
  - All user activity in the systems management console should be logged

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15. Capacity Management Objectives

Capacity Management ensures the timely availability of sufficient infrastructural capacity to process, transport, and store data now and in the future

Objectives:

1. **Optimize Resource Usage –** Ensures CPU, memory, and storage are efficiently utilized.
2. **Prevent Performance Issues –** Avoids system slowdowns due to overutilization.
3. **Cost Efficiency –** Reduces unnecessary hardware expenses.
4. **Scalability –** Helps in planning for future growth.
5. **Proactive Monitoring –** Identifies potential capacity issues before they impact services.

16. Storage Replication Types & Differences

Replication Type	Definition	Key Difference
Synchronous Replication	Data is written to both primary and secondary storage at the same time.	Ensures real-time data consistency but adds latency.
Asynchronous Replication	Data is first written to primary storage, then replicated to secondary storage after a delay.	Faster performance but may cause data loss if failure occurs before replication.

Security Features & How They Work

1. **Firewalls** – Block unauthorized access by checking incoming and outgoing data.
2. **Encryption** – Turns data into unreadable code to keep it safe.
3. **Access Control** – Allows only approved users to access the system.
4. **IDS/IPS** – Detect and stop any suspicious or harmful network activity.
5. **Backup & Recovery** – Saves extra copies of data to restore it after any failure.

How Performance Affects Operating System

Long question

* The performance of an operating system is dependent on:
  * The performance of the underlying hardware
  * The type of load generated by the applications
  * The configuration of the operating system itself
* Some operating system performance can be gained by:
  * Increasing memory
  * Decreasing the kernel size

19. IT Infrastructure Process

IT infrastructure process includes:

- **Planning & Design –** Setting up hardware, network, and security.
- **Deployment –** Installing servers, networks, and storage.
- **Monitoring & Maintenance –** Ensuring performance, security, and scalability.
- **Upgrades & Optimization –** Improving infrastructure based on usage.

RAID

Short +mcqs

- RAID can be implemented in several configurations, called RAID levels
- In practice, five RAID levels are implemented most often:
  - RAID 0 - Striping
  - RAID 1 - Mirroring
  - RAID 10 - Striping and Mirroring
  - RAID 5 - Striping with distributed parity
  - RAID 6 - Striping with distributed double parity

21. Security Concern Process

- **Risk Assessment –** Identifying security threats.
- **Access Control –** Managing user permissions.
- **Data Encryption –** Protecting data from unauthorized access.
- **Incident Response –** Handling breaches and security issues.

24. Network Layer Protocol & Layers

The **Network Layer** is responsible for routing and delivering data between devices across different networks.

Protocols in Network Layer:

- **IP (Internet Protocol)** – Responsible for addressing and routing packets.
- **ICMP (Internet Control Message Protocol) –** Used for diagnostics (e.g., Ping command).
- **ARP (Address Resolution Protocol) –** Maps IP addresses to MAC addresses.

Layers in Network Model:

1. **Physical Layer –** Deals with transmission of raw data bits.
2. **Data Link Layer –** Manages MAC addressing and error detection.
3. **Network Layer –** Handles IP addressing and routing.
4. **Transport Layer –** Ensures end-to-end communication (TCP/UDP).
5. **Session Layer –** Manages sessions between applications.
6. **Presentation Layer –** Data encryption and formatting.
7. **Application Layer –** Provides network services to applications (HTTP, FTP, etc.).

Vendor Bidding means choosing the best company (vendor) to buy IT items like servers, networks, etc.

Steps:

1. **RFP (Request for Proposal):**
   Company asks vendors to send their offers.
2. **Evaluate Proposals:**
   Company checks price, features, and if it fits their system.
3. **Negotiation:**
   They discuss and finalize price and terms.
4. **Selection:**
   Company picks the best vendor and buys the items.

🔷 **Vendor Lock-in Risk:**
Some vendors use special systems, so later it becomes hard to switch to another vendor.

Let me know if you want this even shorter

27. Meaning of RADIUS

RADIUS (Remote Authentication Dial-In User Service) is a networking protocol used for centralized authentication, authorization, and accounting (AAA) of users accessing a network.

□ **Uses:**

- Provides authentication for network devices (e.g., routers, VPNs).
- Manages user permissions and access control.
- Used in enterprise and ISP environments for secure logins.

28. IT Infrastructure's 3 Steps According to Ethics

Privacy and Data Protection • Transparency and Accountability • Accessibility

Deployment Infrastructure Model Names

- Infrastructure can be hosted:
  - On-premises
  - In a colocation
  - Outsourced

DTAP environments

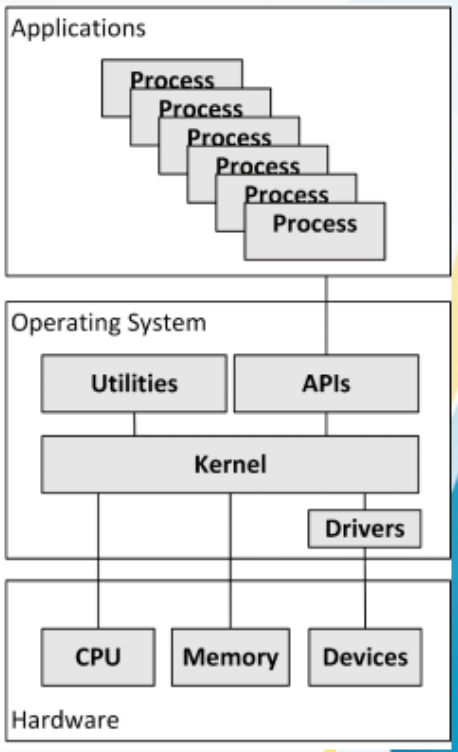
- DTAP:
  - Development
    - New software is developed or existing software is modified
  - Test
    - Software is tested by independent testers
  - Acceptance
    - Software is accepted by a delegation of the user population
  - Production
    - When all tests are successful, the software is deployed in the production environment
- DTAP environments are used in the software development process

- Operating systems provide an abstraction layer between (virtualized or physical) hardware and software applications
- As a service to applications, low level hardware management is handled by the operating system
  - Process management
  - Memory management
  - Interrupt handling
  - Multi user management
  - File locking and file sharing

Long question

Operating System Building Blocks

- The kernel is the heart of an operating system
  - Starts and stops programs
  - Manages the file system
  - Performs low level tasks that most programs need
    - The kernel schedules access to hardware to avoid conflicts if two programs try to access the same resource or device simultaneously



33. Two File Transfer Protocols

1. **FTP (File Transfer Protocol):** Used for transferring files over a network.
2. **SFTP (Secure File Transfer Protocol):** Secure version of FTP using SSH encryption.

Cloud computing

Cloud computing = Internet se IT services lena, jab chaho, jitna chaho, sirf utna paisa do.

- Cloud computing is one of the most important paradigm shifts in computing in recent years
- Definition NIST:

*Cloud computing is a model for enabling ubiquitous, convenient, **on-demand network access** to a **shared pool** of configurable computing resources that can be **rapidly provisioned** and released with **minimal management effort** or service provider interaction*

- Cloud computing is an outsourcing model, in which IT services are provided and paid based on actual on-demand use

Cloud deployment models

- A public cloud is accessible through the internet, and available to the general public
- A private cloud is operated solely for a single organization, whether managed internally or by a third-party, and hosted either on premises or external
- A community cloud is much like a private cloud, but shared with a community of organizations that have shared concerns
- In a hybrid cloud deployment, a service or application is provided by a combination of a public cloud, and a community cloud and/or a private cloud